

NSL_Dataset

duration	Length of the network connection in seconds	
protocol_type	Transport protocol used (TCP, UDP, ICMP)	
service	Network service on the destination (e.g., HTTP, FTP, SMTP)	
flag	Status of the connection (normal, error, rejected, etc.)	
src_bytes	Number of bytes sent from source to destination	
dst_bytes	Number of bytes sent from destination to source	
land	1 if source and destination IP and port are the same, else 0	
wrong_fragment	Number of incorrect IP fragments	
urgent	Number of urgent packets in the connection	
hot	Number of suspicious system commands	
num_failed_logins	Number of failed login attempts	
logged_in	1 if login was successful, else 0	
num_compromised	Number of compromised conditions detected	
root_shell	1 if root shell was obtained, else 0	
su_attempted	1 if "su root" command was attempted, else 0	
num_root	Number of root-level accesses	
num_file_creations	Number of file creation operations	
num_shells	Number of shell prompts invoked	
num_access_files	Number of operations on access control files	
num_outbound_cmds	Number of outbound commands in an FTP session (always 0 in NSL-KDD)	
is_host_login	1 if login belongs to the host list, else 0	
is_guest_login	1 if login is a guest login, else 0	
count	Number of connections to the same host in the last 2 seconds	
srv_count	Number of connections to the same service in the last 2 seconds	
serror_rate	Percentage of connections with SYN errors	
srv_serror_rate	SYN error rate for the same service	
rerror_rate	Percentage of connections with REJ errors	
srv_rerror_rate	REJ error rate for the same service	
same_srv_rate	Percentage of connections to the same service	
diff_srv_rate	Percentage of connections to different services	
srv_diff_host_rate	Percentage of connections to different hosts for the same service	

dst_host_count	Number of connections to the destination host	
dst_host_srv_count	Number of connections to the destination host using the same service	
dst_host_same_srv_rate	Rate of connections to the same service on the destination host	
dst_host_diff_srv_rate	Rate of connections to different services on the destination host	
dst_host_same_src_port_rate	Rate of connections from the same source port	
dst_host_srv_diff_host_rate	Rate of connections to different hosts for the same service	
dst_host_serror_rate	SYN error rate for the destination host	
dst_host_srv_serror_rate	SYN error rate for the destination host's service	
dst_host_rerror_rate	REJ error rate for the destination host	
dst_host_srv_rerror_rate	REJ error rate for the destination host's service	
label	Type of traffic (normal or specific attack)	
difficulty_level	Difficulty score assigned to the record (not used in training)	